llnl_c1/lulesh · lulesh.cc · llm_pragmas_v5_exact · claude-haiku-4-5-20251001 · FASTER · RIGHT = attempt 1 of 87, rejected by the gate at 'correctness'

−1 / +61 lines of 1967 · 1 parallel region(s) added · 29 of 87 candidate(s) kept · RESTRUCTURING (45 code line(s) added, 1 removed) · restructuring level 0

BEFORE

```
1960  #ifdef SEDOV_SYNC_POS_VEL_LATE
1961  #endif
1962
1963     LagrangeElements(domain, domain.numElem());
1964
1965     CalcTimeConstraintsForElems(domain);
1966
1967  }
```

ATTEMPT 1 OF 87, REJECTED BY THE GATE AT 'CORRECTNESS'

file

```
1960  #ifdef SEDOV_SYNC_POS_VEL_LATE
1961  #endif
1962
1963     LagrangeElements(domain, domain.numElem());
1964
1965     /* Parallelize time constraints calculation loop over regions */
1966     Real_t dtcourant = Real_t(1.0e+20);
1967     Real_t dthydro = Real_t(1.0e+20);
1968
1969  #pragma omp parallel for collapse(1) reduction(min:dtcourant,dthydro) \
1970      schedule(static) shared(domain)
1971     for (Index_t r = 0; r < domain.numReg(); ++r) {
1972        /* Courant constraint computation for region r */
1973        Real_t qqc2 = Real_t(64.0) * domain.qqc() * domain.qqc();
1974        Real_t dtcourant_tmp = Real_t(1.0e+20);
1975        Index_t courant_elem = -1;
1976
1977        for (Index_t i = 0; i < domain.regElemSize(r); ++i) {
1978           Index_t indx = domain.regElemlist(r, i);
1979           Real_t dtf = domain.ss(indx) * domain.ss(indx);
1980
1981           if (domain.vdov(indx) < Real_t(0.)) {
1982              dtf = dtf
1983                  + qqc2 * domain.arealg(indx) * domain.arealg(indx)
1984                  * domain.vdov(indx) * domain.vdov(indx);
1985           }
1986
1987           dtf = SQRT(dtf);
1988           dtf = domain.arealg(indx) / dtf;
1989
1990           if (domain.vdov(indx) != Real_t(0.)) {
1991              if (dtf < dtcourant_tmp) {
1992                 dtcourant_tmp = dtf;
1993                 courant_elem = indx;
1994              }
1995           }
1996        }
1997
1998        if (courant_elem != -1) {
1999           dtcourant = dtcourant_tmp;
2000        }
2001
2002        /* Hydro constraint computation for region r */
2003        Real_t dthydro_tmp = Real_t(1.0e+20);
2004        Index_t hydro_elem = -1;
2005
2006        for (Index_t i = 0; i < domain.regElemSize(r); ++i) {
2007           Index_t indx = domain.regElemlist(r, i);
2008
2009           if (domain.vdov(indx) != Real_t(0.)) {
2010              Real_t dtdvov = domain.dvovmax() / (FABS(domain.vdov(indx)) + Real
2011
2012              if (dthydro_tmp > dtdvov) {
2013                 dthydro_tmp = dtdvov;
2014                 hydro_elem = indx;
2015              }
2016           }
2017        }
2018
2019        if (hydro_elem != -1) {
2020           dthydro = dthydro_tmp;
2021        }
2022     }
2023
2024     domain.dtcourant() = dtcourant;
2025     domain.dthydro() = dthydro;
2026
2027  }
```

red = removed green = added yellow = parallel region (#pragma omp) right strip = the whole file: green where changed, yellow where a parallel region now is